Trenitalia
ETR1000 VZI-6

Coupler

Revision	Author	Date	Revision Description
00.1	R. Schiavo	27/09/2023	Draft Release Generated from DOORS module /Platform.-.Zefiro-Europe/../2F_02.-.Exterior Fittings-VF.-.Zefiro-Europe/2F_02.04.-.Coupler-VF.-.Zefiro-Europe/Coupler-VF.CONCEPT, current 5.2 Doc 3ECP413922-0204_
00	R. Schiavo	27/09/2023	First Emission
1.1	R. Schiavo	13/10/2023	#.HR01 and #.HR02 requirements have benn added, the SIL assigned to #.304 req has been modified
1.0	R. Schiavo	13/10/2023	Emission
2.1	R. Schiavo	29/04/2024	Added Redundance IO as reported in FA010024082 rev 01 Modified Requirements: 2F_02.04.Zefiro-Europe.CONCEPT.304,.171,.412,.167
2.0	R. Schiavo	30/04/2024	Emission – BL1 baseline
3.01	G. Spina	01/06/2025	Added req HR03, HR04, HR05 Modified HR01
3.0	G. Spina	06/08/2025	Emission – BL4 baseline

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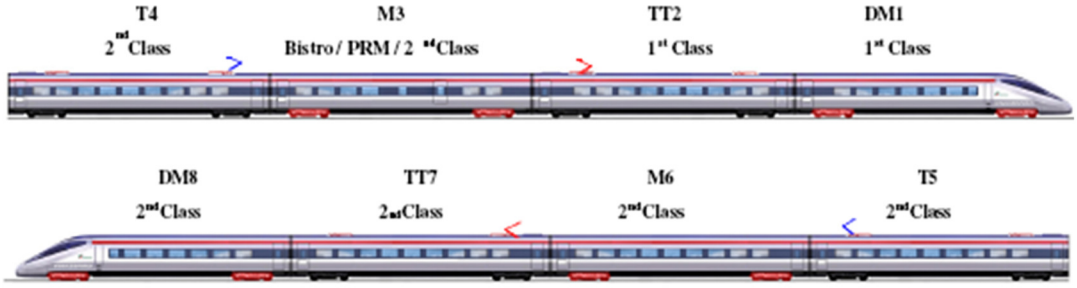
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1 Introduction

This document is a function specification for the control of the *Coupler* and *Coupler Covers* system for ETR1000 / VZI6 for customer Trenitalia.

1.1 Train Configuration

Each ETR1000 / VZI6 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
IPB	Illuminated Push Button
TCMS	Train Control and Management System

2 System Overview

2.1 System Configuration

The SFRS module coupler consists of two separate systems:

Coupler: Connects two train sets with automatic coupler mechanically, electrically and pneumatically.

Coupler covers: Covers the coupler in single traction and opens front with automatic coupler for double traction.

Coupling sequence is activated by a control device on driver's desk as described in SFRS Conventional Train Control. The coupler covers can be opened via buttons on the IDU. Decoupling is activated by a control device on the driver's desk. The coupler covers are closed by a push button on the IDU and automatically at a certain speed.

The train level function of coupling and decoupling are described in the SFRS conventional train control [2]. The requirements for the coupling control are described within this document.

3 Functional requirements

3.1 Coupling

3.1.1 Function Design

Coupling is a process to create a mechanical and electrical couple of two trains. By this, pneumatic and electric signals can be transferred between both parts. Coupling is also possible between a train and a loco, whereas no electrical coupling can be done in this case.

The coupling function requires the activation of the "Couple Speed Mode" for which all the conditions are described in SFRS conventional train control [2]. Herein, also the switch to access this mode is described. When activating this mode the coupler cover in the front will open automatically. By this coupler mode also a speed limit is given which ensures the automatic coupling without damaging the coupler. Moreover, the coupler cover can be opened via the IDU below a certain speed.

Following signals are needed to define the status of coupling condition:

-Mechanical coupled/uncoupled from mechanical switch

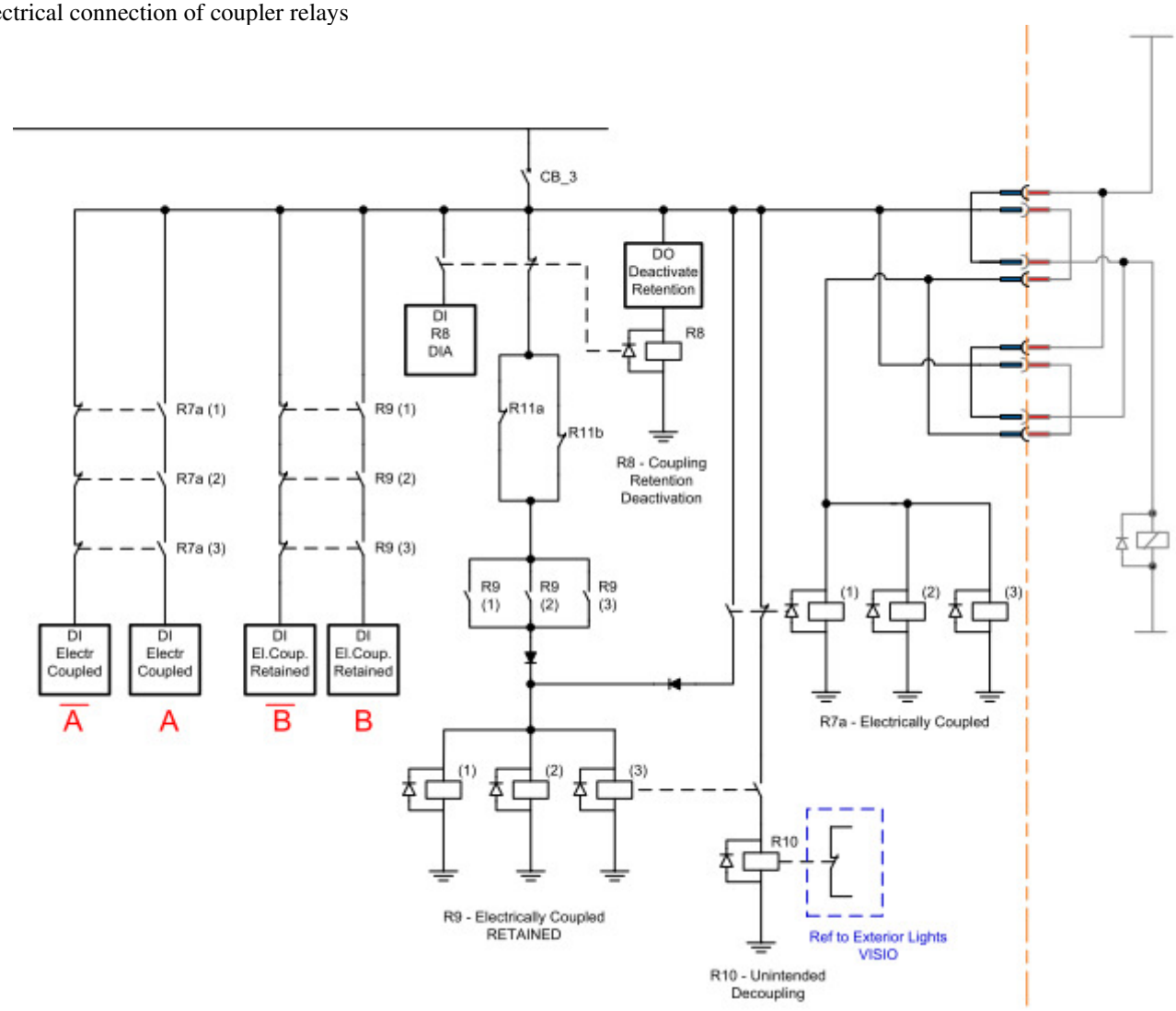
-Electrical coupled indicated by "Signal A" which is high in case electrical coupler is fully connected - Relays R7a.

-Electrical coupled indicated by "Signal B" which is high in case electrical coupler was once fully connected and not intended decoupled - Relays R9.

Signal B has a stored function and is used to detect unintended decoupling.

A detailed description of the coupler cover control can be found in chapter 3.3

Electrical connection of coupler relays



Normal condition

In normal couple condition the couple sequence shows no failure related to coupler cover opening. Both mechanical and electrical couple conditions are aligned and Signal A and Signal B for electrical coupling are high.

Failure cases

1. Electrical coupled detected but not coupled
2. Electrical coupled but not detected
3. Mechanical coupled but mechanical coupled signal not set by coupler
4. Not mechanical coupled but mechanical coupled signal set by coupler
5. Implausible electrical coupling Signal A and B

Failure detection

1. When electrical is set and mechanical coupled not set at the same time an event shall be set
2. When electrical is not set and mechanical coupled is set at the same time an event shall be set
3. When mechanical coupled not set and electrical coupled is set at the same time an event shall be set
4. unintended decoupling is result. Electrical coupled Signal A and Signal B used for uncoupling detection
5. electrical coupled Signal A and Signal B used for coupling status detection

Failure handling

1. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
2. No multiple traction possible. Only towing possible.
3. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
4. Emergency brake applied. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.
5. Repeat coupling process max. 2 times. If coupling not successful after 3rd try no multiple traction possible. Unplanned workshop.

3.1.2 Requirements

ID	2F_02.04.Zefiro-Europe.TRS.46
Description	The coupler shall automatically connect two trains together mechanically, pneumatically and electrically and make it possible to operate the whole train from the front-end cab.
Safety level	N/A

ID	2F_02.04.Zefiro-Europe.TRS.42
Description	Emergency brake shall automatically be applied in both trainparts in case of unintended decoupling.
Safety level	SIL 3

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.360	Automatic Coupler shall provide status of coupling device via microswitches installed on coupled head: Mechanical Coupled (Switch S14) Mechanical UNCoupled (Switch S15)	Derived Requirement	4S_02.06.-.Couplers		N/A
2F_02.04.Zefiro-Europe.CONCEPT.301	The signal of the mechanical coupling switch (S14) and mechanical uncoupled switch (S15) shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.302	TCMS shall acquire following status S14 = 1 -> Mechanically coupled switch set (acquired using K2 delayed opening relay)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.303	TCMS shall acquire following status S15 = 1 -> Mechanically Uncoupled switch set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

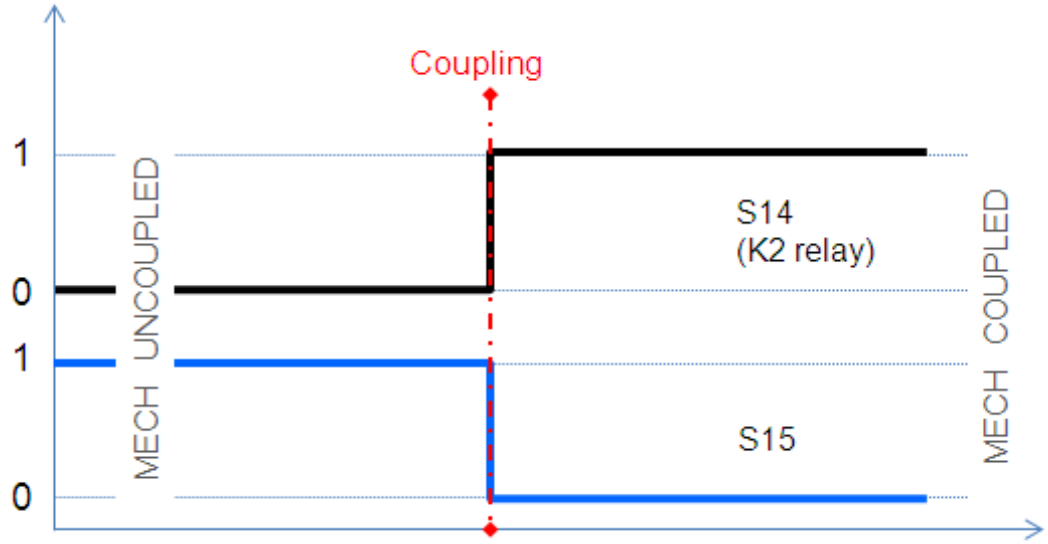
There are two signals “Signal A” and “Signal B” for defining the electrical coupled condition.
"Signal A" shall be derived from series of all R7a relays NO contacts
"Signal not-A" shall be derived from series of all R7a relays NC contacts

"Signal B" shall be derived from series of all R9 relays NO contacts
"Signal not-B" shall be derived from series of all R9 relays NC contacts

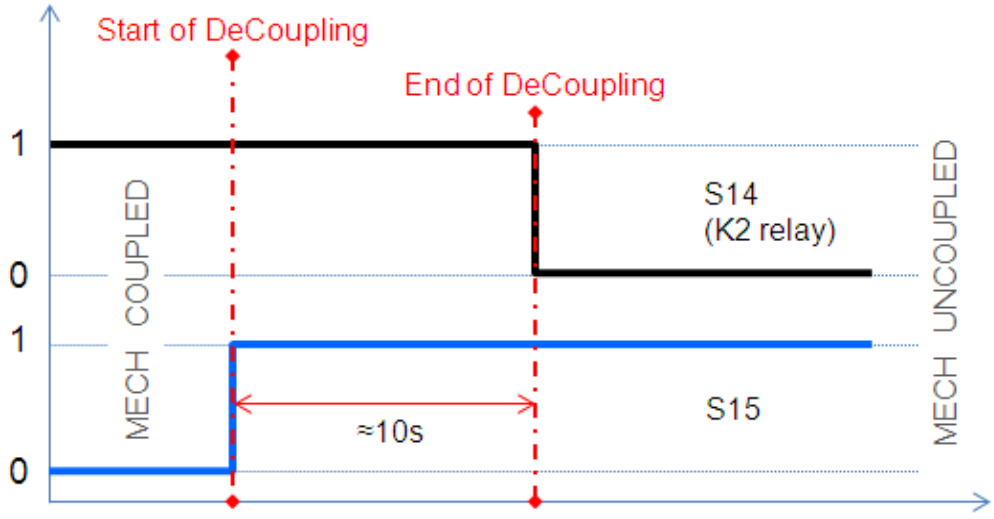
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.304	Relays detecting the electrical coupling connection (R7a) shall be installed. The series of NO contacts of all R7a relays shall be connected to redoundant safety inputs of TCMS The series of NC contacts of all R7a relays shall be connected to redoundant safety inputs of TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_02.04.Zefiro-Europe.CONCEPT.305	Relays retaining the electrical coupling connection (R9) shall be installed (relay status shall be mainained after the first connection untill a dedicated reset is done). The series of NO contacts of all R9 relays shall be connected to redundant inputs of TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

	The series of NC contacts of all R9 relays shall be connected to redundant inputs of TCMS				

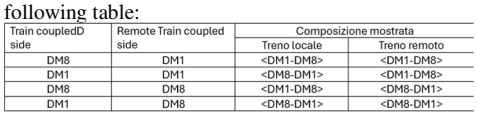
Mechanical Coupling sequence signals



Mechanical DeCoupling sequence signals



Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.307	Following logic shall be used to define the Train couple condition: S14=1 & S15=0 → Train Mechanically coupled (mc) S14=0 & S15=1 → Train Mechanically UNcoupled (muc) A=1 & B=1 → Train Electrically Coupled (ec) A=0 & B=0 → Train Electrically UNcoupled (euc) A=0 & B=1 → Unintended decoupling (T1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	(mc) & (ec) → Train coupled multiple operation (T3) (mc) & (euc) → Train mechanically coupled only (T5) (muc) & (euc) → Train uncoupled (T6)				
2F_02.04.Zefiro-Europe.CONCEPT.207	In case of Unintended decoupling (T1) an alarm shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.364	For coupling conditions T3 and T5 an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.308	Following logic shall be used to define the faulty status of couple conditions: A=1 & B=0 → Faulty coupling retained logic (T2) (muc) & (ec) → No mechanical coupling detected (only electrical) (T4)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.309	In case of train couple failure condition are set for more than 15s, T2 - unexpected electrical Failure T4 - unexpected electrical Failure an event shall be generated for every case above	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.363	In case mechanical coupling status cannot be detected → S14 = S15 for more than 15s (or both signals are not valid) OR Mechanical coupling status changes at speed (if v > 3km/h) Following function shall be inhibited – Coupler Covers Automatic closing if v>10 km/h (only in the cab where Coupling status cannot be detected) – Remote decouplig that affects automatic coupler in the cab where Coupling status cannot be detected An event shall be set to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.355	"Signal A" and "Signal not-A" shall always have opposite values. In case of incoherence an event shall be set suggesting a maintenance check on coupling relays	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.356	"Signal B" and "Signal not-B" shall always have opposite values. In case of incoherence an event shall be set suggesting a maintenance check on coupling relays	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.HR05	In case of failure configuration failure “Signal A” and “signal B” , AND remote train provide safety information of coupled status, the TCMS shall consider the status of remote train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_02.04.Zefiro-Europe.CONCEPT.310	Train coupled configuration shall be displayed in IDU menus	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.HR01	When the coupling or decoupling is detected the new train configuration shall be displayed with train is standstill. On the train configuration menu the new configuration to confirm shall be presented.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2
2F_02.04.Zefiro-Europe.CONCEPT.HR02	If the safe acknowledge button is pressed the new configuration shown on IDU shall be accepted.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2
2F_02.04.Zefiro-Europe.CONCEPT.HR03	The train configuration shall report the position of enabled cab and train orientation based on the status of coupling relay received from remote train	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.04.Zefiro-Europe.CONCEPT.HR04	The train configuration shall be set in according following table: 	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

Train coupledD side	Remote Train coupled side	Composizione mostrata	
		Treno locale	Treno remoto
DM8	DM1	<DM1-DM8>	<DM1-DM8>
DM1	DM1	<DM8-DM1>	<DM1-DM8>
DM8	DM8	<DM1-DM8>	<DM8-DM1>
DM1	DM8	<DM8-DM1>	<DM8-DM1>

2F_02.04.Zefiro-Europe.CONCEPT.349	After an "unintended decoupling" is detected, TCMS shall reset "Signal B" ("Electrically coupled retained status") by setting a DO, only when the train is in standstill and the driver has acknowledged the new train configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.350	The storage of the "unintended decoupling" signal shall be designed to be available at every possible state of the battery.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.96	The electrical coupled connection shall be verified by dedicated pins in each of the electrical coupler heads to generate signal electrical coupled.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_02.04.Zefiro-Europe.CONCEPT.209	When electrical heads loose contact the emergency brake loop actuator line shall be de-energised in both train sets.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3

3.2 Decoupling

3.2.1 Function Design

The driver is responsible for the operation of the decoupling command. After decoupling the coupler cover closes automatically at a certain speed further described in chapter 3.3. In case of manual cover close command, the driver has to check that there is no obstruction interfering the closure of the coupler cover.

Decoupling is activated by a control device on driver's desk. Depending on the position of a rotary switch the local or remote coupler will be decoupled. The two decoupling magnet valves EVCD 02.14 and EVCD2 02.20 are in scope of brake system.

When decoupling order is received and the train has applied parking brakes the coupler will be decoupled pneumatically.

The supervision of the uncouple condition is described in the couple chapter 3.1.

Failure condition

1. Decoupling order is given but coupler is not decoupling
2. Decoupling order is not given but coupler is decoupling

Failure detection

1. The train coupler condition is supervised by electrical and mechanical switches, refer to 3.1
2. Detected by unintended decoupling, refer to 3.1

Failure handling / effect

1. manual decoupling by authorised stuff. No impact on train operation.
2. emergency brake activated

3.2.1 Requirements

ID	2F_02.04.Zefiro-Europe.TRS.77
Description	The driver shall be able to order the two trainsets to decouple by a control in the cab.
Safety level	SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.78
Description	Decoupling of two trainsets shall only be possible from the active cab. It shall be possible to disconnect the two trainsets in front of the active cab (if a cab in the middle of the 16 cars train is activated) or on the opposite side in remote (if a cab at the end of the 16 cars is activated and the disconnection is done in the middle).
Safety level	SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.79
Description	Decoupling shall only be possible if the train is standing still.
Safety level	SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.49
Description	Two trainsets shall be decoupled when a decoupling order from the cab is received.
Safety level	SIL 0

For both, remote and local decoupling, the valves shall be energized for 10s.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.312	There shall be a decoupling control device on the driver's desk.	Derived Requirement	4S_08.07.01-.Drivers Desk		N/A
2F_02.04.Zefiro-Europe.CONCEPT.313	The different positions of the decoupling control device • Local • 0 • Romote shall be connected to TCMS digital inputs	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.320	When remote and local decoupling are requested both decoupling valves shall be energized.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits 4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.321	To ensure a successful decoupling, the "mechanically coupled" relay shall be designed as a time relay maintaining its active state for 10 seconds. The relay shall be controlled by the coupler switch S14 "mechanically coupled"	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.171	TCMS redundant outputs for remote and local decoupling shall be connected to the two decoupling valves for coupler.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.163	When remote de-coupling sequence is started TCMS shall energize the decoupling valves for coupler in the rear end of the train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.388	When local de-coupling sequence is started TCMS shall energize the decoupling valves for coupler in the front end of the train	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.374	When any de-coupling command is set (local or remote) TCMS shall check that both decoupling valves are correctly energized. In case one or more feedback signals are not set an event shall be displayed (decoupling not possible)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.375	When there is NO de-coupling command set (local and remote) TCMS shall check that both decoupling valves are correctly de-energized. In case one or more feedback signals are unduly set an event shall be displayed (isolate decoupling pipe to avoid unwanted decoupling)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.347	In case of decoupling commanded by TCMS the valves shall be energized for 10s.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_02.04.Zefiro-Europe.CONCEPT.315	It shall be possible to start the remote and local decoupling only if: • The decoupling command comes from the active cab AND • the Master Controller is in neutral position (coast) AND • the train is at standstill AND • Reverse direction is NOT selected AND • min 75% of parking brakes are applied in each train of the consist individually	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.316	An advisory event shall be set in case of remote decoupling is not allowed reporting the missing condition	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.317	Decoupling shall be considered as successful if following conditions are true: – Train was previously in a valid coupled status → (T3) or (T5) – Train coupled condition becomes "uncoupled" → (T6)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.318	If the decoupling is commanded, Signal B (EI coupled retained status) shall be restored de-energizing R9 relays	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.319	There shall be the possibilty to reset the stored Signal B de-energizing R9 relays, by a TCMS output	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.80
Description	After uncoupling, the decoupled train part shall apply emergency brake. In the part with active cab the new configuration of the security loops shall be acknowledged by the driver.
Safety level	SIL 0

Brake condition of a train with deactivated cab and not applied parking brake can be found in Brake SFRS.

ID	2F_02.04.Zefiro-Europe.TRS.81
Description	Failure of the decoupling sequence shall be indicated to the driver.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.323	When decoupling is requested and "mechanically decoupled" signal is not present after 30s and Event shall be generated and shown on IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_02.04.Zefiro-Europe.TRS.84
Description	Driver shall be able to electrically uncoupling the Multiple Unit
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.397	**GIVEN** (TCMS active) **WHEN** (TCMS detects that the pneumatic cock which enable / disable the e-uncoupler Command is open (0 V) for at least 5 s) **THEN** (TCMS shall set a warning message to the driver)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_02.04.Zefiro-Europe.CONCEPT.398	**GIVEN** (TCMS active) **WHEN** (TCMS detects that the pneumatic cock which enable / disable the e-uncoupler Command is closed (110 V) for at least 5 s) **THEN** (TCMS shall reset a warning message to the driver)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

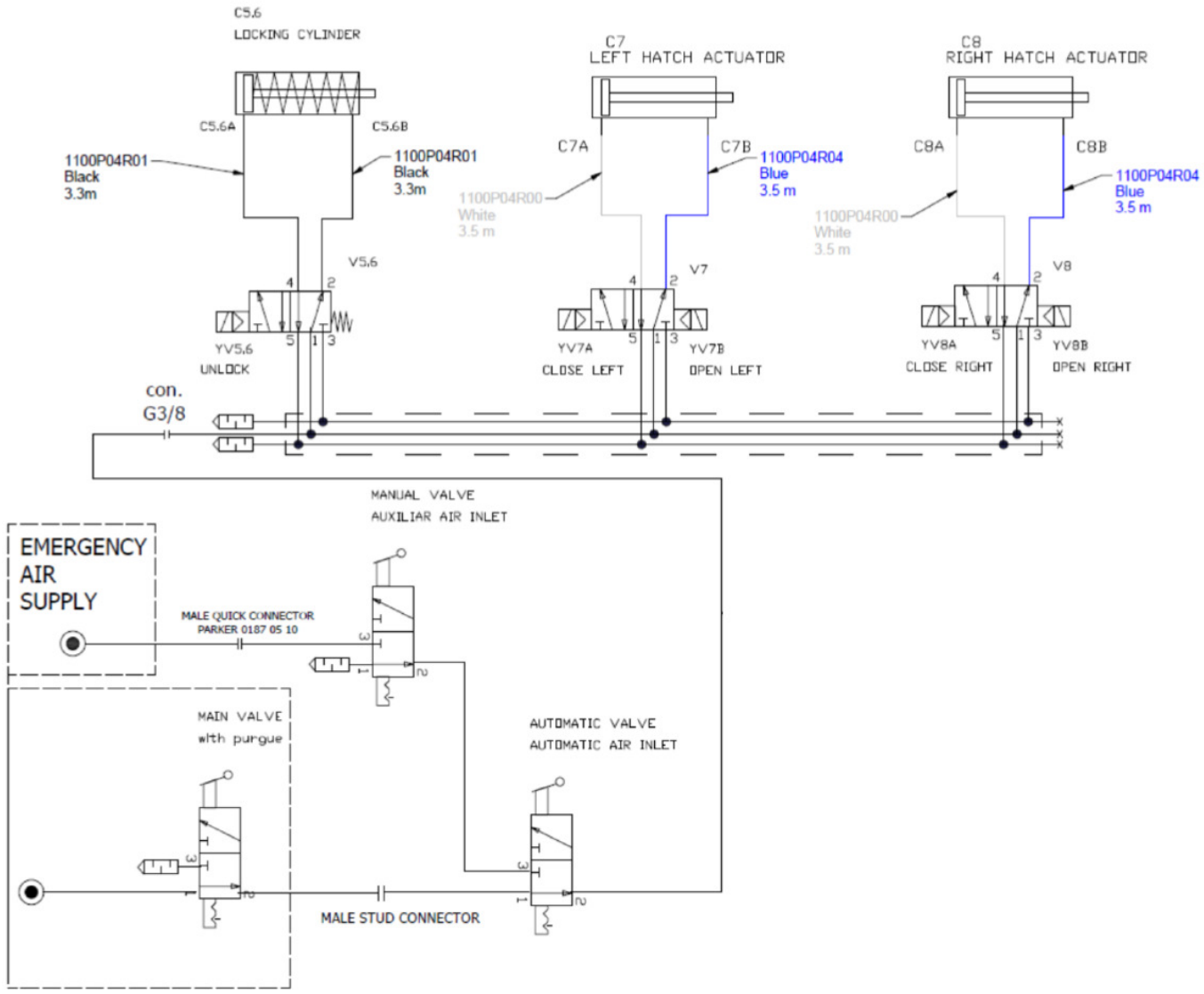
3.3 Control of Coupler Cover

3.3.1 Function Design

The coupler covers of any end of the train set can be opened/closed from an active cab. The opening of coupler cover can be done in two ways. On one hand, the coupler cover in the front is opened automatically after the activation of couple speed mode. On the other hand, the cover can be opened by using forcing it via IDU. This is possible only below a certain speed. The cover will start to close automatically after exceeding a speed of 10km/h. Moreover, the cover can be closed via IDU button. The driver then is responsible the check the coupler environment and has to ensure no obstruction can collide with the cover.

Control of coupling cover and supervision if any failure in the opening or closing sequence is present are done by separate solenoid valves and position switches. The opening/closing sequence, below explained using the example of opening, consists of three steps.

1. Both the right and left cover hatch are commanded to be unlocked by energising the solenoid valve for unlocking/locking (V5.6). The corresponding switch detects the unlocking.
2. Both the right and left cover hatches are commanded to open by separate solenoid valves (V7 and V8) feeding a pressure cylinder. Final position switches detect the successfully opened hatches.
3. Finally, both hatches are locked by de-energising the solenoid valve (V5.6). Again, the locked status is detected by the switch.



Normal conditions

In normal condition the coupler cover in the front of the driving direction opens as soon as the Coupler Speed Mode is activated. The cover is locked after the open position is detected. For closing the cover, first the hatches are unlocked and then are closed completely. After closure is detected, the hatches are locked again.

Failure cases

1. coupler covers do not open automatically after activation of coupler speed mode
2. one or both coupler cover are not opening / closing when pressing IDU coupler cover control buttons
3. coupler covers closing without order
4. coupler covers are closed/ opened but not locked
5. Implausible indication of locked/unlocked or open/close
6. Coupler covers remain open at speed greater than 50km/h

Failure detection

1. Coupler status supervised by final position sensor and indicated to the driver
2. Coupler status supervised by final position sensor and indicated to the driver
3. Coupler status supervised by final position sensor and indicated to the driver
4. Coupler status (including locked/unlocked) supervised by final position sensor and indicated to the driver

5. if there is a implausible indication, a failure event will be created
6. Coupler status supervised by final position sensor and indicated to the driver

Failure handling

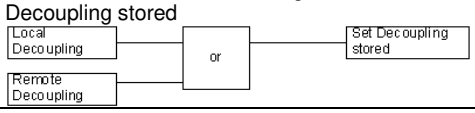
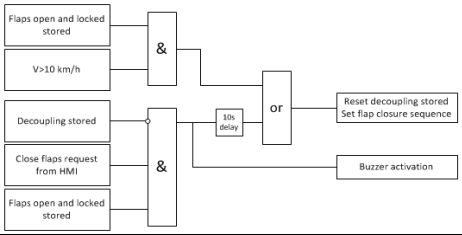
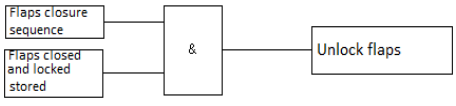
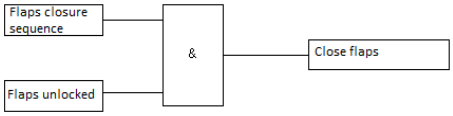
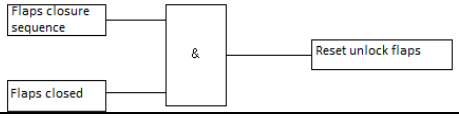
1. Opening of cover can be command by IDU as a second path. If this won't work either, the cover can be opened manually.
2. Covers must be opened manually.
3. Opening must be commanded again. If failure remains present, cover must be opened manually.
4. Pneumatic path must be vented. Spring loaded cover lock will close automatically.
5. If status of coupler cover is unclear, it must be controlled manually
6. Speed limit of 140km/h is set.

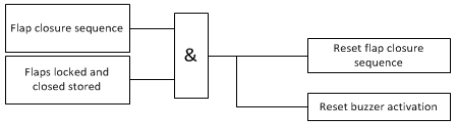
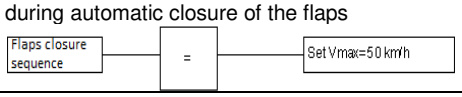
3.3.2 Requirements

ID	2F_02.04.Zefiro-Europe.TRS.66
Description	The driver shall be able to order the coupler cover to close by a control in the drivers desk after the train has been decoupled.
Safety level	N/A

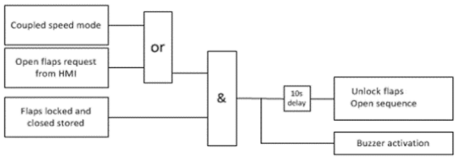
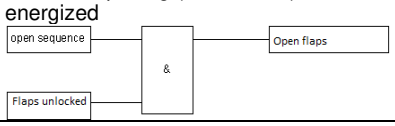
ID	2F_02.04.Zefiro-Europe.TRS.63
Description	The opening of the coupler covers shall be interlocked at speeds >10 km/h.
Safety level	SIL 0

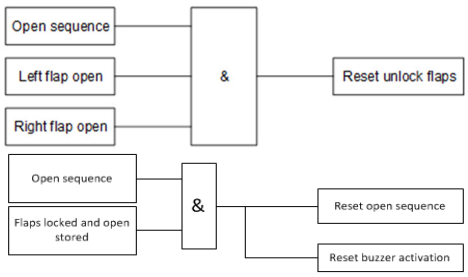
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.405	TCMS shall acquire following signals to define coupler covers status Sensor S5 → LEFT hatch CLOSED Sensor S6 → LEFT hatch OPEN Sensor S7 → RIGHT hatch CLOSED Sensor S8 → RIGHT hatch OPEN Sensor S11.9 → FRONT hatch LOCKED Sensor S12.10 → FRONT hatch UNLOCKED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.406	TCMS shall command control valves for front cover opening/closing/locking V7A/V8A → Front hatches CLOSING command V7B/V8B → Front hatches OPENING command V5.6 → Front hatches UNLOCK command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.338	The status of **OPEN/CLOSED** coupler flaps shall be stored within TCMS according following logic: S5=0 & S6=1 → **LEFT flap OPEN** S5=1 & S6=0 → **LEFT flap CLOSED** S5 = S6 or any S5, S6 not valid → **LEFT flap position FAULTY** S7=0 & S8=1 → **RIGHT flap OPEN** S7=1 & S8=0 → **RIGHT flap CLOSED** S7 = S8 or any S7, S8 not valid → **RIGHT flap position FAULTY**	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.407	The status of **UNLOCKED/LOCKED** coupler flaps shall be stored within TCMS according following logic: Sensor S11.9 = 1 & Sensor S12.10 = 0 → **flaps LOCKED**	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	Sensor S11.9 = 0 & Sensor S12.10 =1 → **flaps UNLOCKED** Sensor S11.9 = Sensor S12.10 or any S11.9, S12.10 not valid → **flaps locking FAULTY**				
2F_02.04.Zefiro-Europe.CONCEPT.373	In case any cover is detected Faulty (position faulty OR locking faulty) for more than 15s, – automatic closing shall be inhibited	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.371	In case any cover is detected Faulty (position faulty OR locking faulty) for more than 15s, a warning shall be set to the driver to check and override cover position using IDU commands	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.332	If local decoupling or remote decoupling is selected TCMS shall set the signal Decoupling stored 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.182	TCMS shall start the coupler covers closing sequence when following conditions are set: – Train speed is above 10 km/h – The train has just decoupled and accelerates from that condition – Local automatic coupler status is NOT FAULTY – Local automatic coupler status is MECHANICALLY UNCOUPLED – Local cover status is OPEN and LOCKED (both left and right flap) – Local cover status is NOT FAULTY (both left and right flap) – NO UNINTENDED DE-COUPLING has occurred The closure is requested by soft key and blocked in case the train has decoupled and not yet moved away from the decoupled train 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.408	The closure sequence shall unlock the flaps,  close the flaps  and lock the flaps when closed 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	and reset the flaps closure and buzzer activation when finished 				
2F_02.04.Zefiro-Europe.CONCEPT.333	TCMS shall set max speed of the train to 50km/h during automatic closure of the flaps 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.47
Description	The driver shall be able to order the coupler covers to open by a control in the drivers desk.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.63	The IDU shall provide a status menu for coupler and coupler covers to show the actual condition in front or rear car end: – Coupler cover opened/closed – Coupler cover locked/unlocked	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.117	For opening/closing the coupler covers at standstill there shall be following buttons on the IDU in the actived cab: for the driver: – Open/close coupler cover at front end of train – Open/close coupler cover at rear end of train – Set flaps locked and closed stored local/ remote (signals from position/locking sensor will be bypassed) – Set flaps open and locked stored local/ remote (signals from position/locking sensor will be bypassed) (It shall be prevented by TCMS software that the signals “flaps open and locked stored” and “flaps closed and locked stored” can be active at the same time. Whenever one is selected, the other shall be reset)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.66	The manual (via IDU) opening/ closing of the front cover shall not be allowed at train speed above 10km/h	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.409	When opening command is sent by TCMS or by IDU button, first the solenoid valve for cover locking (V5.6) is energised 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.410	After completed unlocking, the solenoid valves for cover opening (V7B & V8B) shall be energized 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_02.04.Zefiro-Europe.CONCEPT.411	When coupler cover are detected as open, the solenoid valve for cover locking (V5.6) shall be de-energized 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.412	Redundant TCMS output shall be connected to the unlock coupler cover solenoid valve	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.167	Redundant TCMS output shall be connected to both coupler cover open/close magnet valves. Each has one open and one close solenoid.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.413	Switches for left and right coupler cover status shall be connected to TCMS input: – Coupler covers closed – Coupler covers opened – Coupler covers unlocked – Coupler covers locked	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.119	In case of Automatic Coupler status is detected ad "MECHANICALLY COUPLED" (mc), related front hatches shall be prevented from closing	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_02.04.Zefiro-Europe.TRS.51
Description	Failure of opening or closing of the coupler cover shall be indicated to the driver.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.118	When coupler covers are not opening or closing on order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.161	When coupler covers is not locked or not unlocked on order an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.390	**GIVEN** (Battery ON) **WHEN** (the cover status from one cover differs from the status of the other cover in respect to "*open*"/"*closed*") **THEN** (A Warning message shall be generated to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.391	**GIVEN** (Battery ON) **WHEN** (the cover status from one cover differs from the status of the other cover in respect to "*locked*"/"*unlocked*") **THEN** (A Warning message shall be generated to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.262	When coupler cover status change without order, an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_02.04.Zefiro-Europe.CONCEPT.298	Speed Limitation shall be set to 250km/h if following condition are verified: – Any FRONT Flap OPEN (or in Open&Locked overridden position) AND – Automatic coupler status is MECHANICALLY UNCOUPLED AND – Coupler Covers NOT in Closed&Locked overridden position AND – NO UNINTENDED DE-COUPLING has occurred	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.04.Zefiro-Europe.CONCEPT.346	In case of open covers, over 50 Km/h a alarm will be displayed on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.376	When a manual (via IDU) opening of the front/rear flaps is requested the related buzzer shall be commanded by the TCMS 5 seconds before "open sequence" is set	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.04.Zefiro-Europe.CONCEPT.384	When a manual (via IDU) closing of the front/rear flaps is requested the related buzzer shall be commanded by the TCMS 5 seconds before "closure sequence" is set	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.04.Zefiro-Europe.CONCEPT.385	Buzzer activation command shall be reset if at least one of the following conditions is verified: – Open sequence is completed OR – Closure sequence is completed OR – Stop command control valves for front cover opening/closing/locking	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.04.Zefiro-Europe.CONCEPT.386	Two redundant TCMS digital outputs shall be connected to the buzzer	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.387	As long as buzzer activation commad is set, an "other" event shall be set to mantainer	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.04.Zefiro-Europe.CONCEPT.392	**GIVEN** (IDU ON) **WHEN** (Both Covers status are "*Open*") **THEN** (Icon "*Open*" on IDU shall be activated)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_02.04.Zefiro-Europe.CONCEPT.393	**GIVEN** (IDU ON) **WHEN** (Both Covers status are "*Closed*") **THEN** (Icon "*Closed*" on IDU shall be activated)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_02.04.Zefiro-Europe.CONCEPT.394	**GIVEN** (IDU ON) **WHEN** (Both Covers status are "*LOCKED*") **THEN** (Icon "*LOCKED*" on IDU shall be activated)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_02.04.Zefiro-Europe.CONCEPT.395	**GIVEN** (IDU ON) **WHEN** (At least one Cover status is "*UNLOCKED*") **THEN** (Icon "*UNLOCKED*" on IDU shall be activated)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

3.4 Coupler heating

3.4.1 Function Design

Heating of coupler is activated in winter conditions at a temperature below 8°C.

When mechanical coupled is detected the heating of each connected coupler is reduced to save energy.

For outside temperature signal the sensor of air condition is used to control coupler heating.

Normal condition

In uncoupled condition there are normally 5 heating elements switched on. In coupled condition two of 5 heating elements are switched off of each coupler. The respective heating elements are switch on below 1°C. They are switched off above 8°C. This behaviour is valid for both, coupled and uncoupled condition.

Failure case

1. Heating is not activated when ordered
2. Heating is activated without order

Failure detection

1. no failure detection possible
2. no failure detection possible

failure handling

1. no influence on train operation
2. no influence on train operation

3.4.2 Requirements

ID	2F_02.04.Zefiro-Europe.TRS.68
Description	Heating cartridge in coupler shall be activated by TCMS if the outside temperature is < 1º C. When heating is already activated it shall be deactivated if the outside temperature is > 8º C.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.04.Zefiro-Europe.CONCEPT.101	Heating cartridge in coupler shall be activated by TCMS if the outside temperature is < 1º C. When heating is already activated it shall be deactivated if the outside temperature is > 8º C.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.04.Zefiro-Europe.CONCEPT.155	TCMS output shall be connected to coupler heating cartridge to allow both heating couples to be controlled indipendendly.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.217	When signal mechanical coupled is set 2 of 5 heating elements shall be switched off.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.04.Zefiro-Europe.CONCEPT.330	When decoupled all heating elements shall be switched on	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.124	Close coupler covers right (magnet valve V8A)	L	2F_02.04.Zefiro-Europe.CONCEPT.117
2F_02.04.Zefiro-Europe.CONCEPT.172	Open coupler covers right (magnet valve V8B)	L	2F_02.04.Zefiro-Europe.CONCEPT.117
2F_02.04.Zefiro-Europe.CONCEPT.240	Close coupler covers left (magnet valve V7A)		
2F_02.04.Zefiro-Europe.CONCEPT.241	Open coupler covers left (magnet valve V7B)		
2F_02.04.Zefiro-Europe.CONCEPT.414	Unlock coupler covers (magnet valve V5.6)	L	2F_02.04.Zefiro-Europe.CONCEPT.117 2F_02.04.Zefiro-Europe.CONCEPT.412
2F_02.04.Zefiro-Europe.CONCEPT.216	Coupler heating	L	2F_02.04.Zefiro-Europe.CONCEPT.101 2F_02.04.Zefiro-Europe.CONCEPT.155
2F_02.04.Zefiro-Europe.CONCEPT.218	Coupler heating when coupled	L	2F_02.04.Zefiro-Europe.CONCEPT.217

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.187	Mechanical coupled	L	
2F_02.04.Zefiro-Europe.CONCEPT.194	Mechanical uncoupled	L	

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.30	Electrical coupled	L	
2F_02.04.Zefiro-Europe.CONCEPT.125	Electrical coupled stored	L	
2F_02.04.Zefiro-Europe.CONCEPT.188	Mechanical coupled	L	
2F_02.04.Zefiro-Europe.CONCEPT.195	Mechanical uncoupled	L	
2F_02.04.Zefiro-Europe.CONCEPT.126	Coupler cover status open right S8	L	2F_02.04.Zefiro-Europe.CONCEPT.118 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.243	Coupler cover status open left S6		2F_02.04.Zefiro-Europe.CONCEPT.262

2F_02.04.Zefiro-Europe.CONCEPT.244	Coupler cover status close right S7		2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.127	Coupler cover status close left S5	L	2F_02.04.Zefiro-Europe.CONCEPT.118 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.415	Coupler cover status unlock S12.10	L	2F_02.04.Zefiro-Europe.CONCEPT.161 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.416	Coupler cover status lock S11.9	L	2F_02.04.Zefiro-Europe.CONCEPT.161 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.189	Heating active	L	

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.31	Decoupling valve V4 (Brake scope 02.14/02)	L	2F_02.04.Zefiro-Europe.CONCEPT.163 2F_02.04.Zefiro-Europe.CONCEPT.171
2F_02.04.Zefiro-Europe.CONCEPT.250	Decoupling valve (stop valve) V0 (Brake scope 02.14/01)		2F_02.04.Zefiro-Europe.CONCEPT.171
2F_02.04.Zefiro-Europe.CONCEPT.210	Electrical coupled stored	L	
2F_02.04.Zefiro-Europe.CONCEPT.131	Close coupler covers right (magnet valve V8A)	L	2F_02.04.Zefiro-Europe.CONCEPT.117 2F_02.04.Zefiro-Europe.CONCEPT.167 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.251	Open coupler covers right (magnet valve V8B)		2F_02.04.Zefiro-Europe.CONCEPT.167 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.132	Close coupler covers left (magnet valve V7A)	L	2F_02.04.Zefiro-Europe.CONCEPT.117 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.252	Open coupler covers left (magnet valve V7B)		2F_02.04.Zefiro-Europe.CONCEPT.167 2F_02.04.Zefiro-Europe.CONCEPT.262
2F_02.04.Zefiro-Europe.CONCEPT.417	Unlock coupler covers (magnet valve V5.6)	L	2F_02.04.Zefiro-Europe.CONCEPT.161 2F_02.04.Zefiro-Europe.CONCEPT.262 2F_02.04.Zefiro-Europe.CONCEPT.412
2F_02.04.Zefiro-Europe.CONCEPT.133	Activate/deactivate heating of coupler	L	2F_02.04.Zefiro-Europe.CONCEPT.101

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.32	Not applicable		

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.04.Zefiro-Europe.CONCEPT.33	Not applicable		

5 References

[1] FA020023100 SFRS Brake control
[2] FA020023107 SFRS Conventional train control
[3] P1101 / 23004965410 Circuit Diagram Coupler
[4] ELE-L007 Electric Diagram Coupler Covers